

DSA / Algorithms (Abdul Bari) - Day 11

45-day plan mapped to the Algorithms playlist

Master Theorem (Dividing) + Root

Master theorem for dividing functions and root-function recurrences.

Watch these Abdul Bari videos

- > 2.4.1 Masters Theorem in Algorithms for Dividing Function #1
- > 2.4.2 Examples for Master Theorem #2
- > 2.5 Root function (Recurrence Relation)

Key concepts

- Master theorem: compare $n^{\log_b a}$ with $f(n)$.
- Three cases decide the result.
- Root-function recurrences via substitution.

Key idea

$T(n) = aT(n/b) + f(n)$: compare $f(n)$ to $n^{(\log_b a)}$.

Tasks (1.5h)

- ☐ Watch the 3 videos
- ☐ Apply the master theorem to 2 examples
- ☐ Solve a root-function recurrence

Playlist: youtube.com/playlist?list=PLDN4rrl48XKpZkf03iYFl-029szjTrs_0